BioOne COMPLETE

A Genome Size for the Appalachian Gametophyte

Authors: Pelosi, Jessie A., Zumwalde, Bethany A., Sorojsrisom, Elissa S., and Sessa, Emily B.

Source: American Fern Journal, 113(4) : 257-262

Published By: The American Fern Society

URL: https://doi.org/10.1640/0002-8444-113.4.257

BioOne Complete (complete.BioOne.org) is a full-text database of 200 subscribed and open-access titles in the biological, ecological, and environmental sciences published by nonprofit societies, associations, museums, institutions, and presses.

Your use of this PDF, the BioOne Complete website, and all posted and associated content indicates your acceptance of BioOne's Terms of Use, available at www.bioone.org/terms-of-use.

Usage of BioOne Complete content is strictly limited to personal, educational, and non - commercial use. Commercial inquiries or rights and permissions requests should be directed to the individual publisher as copyright holder.

BioOne sees sustainable scholarly publishing as an inherently collaborative enterprise connecting authors, nonprofit publishers, academic institutions, research libraries, and research funders in the common goal of maximizing access to critical research.

Downloaded From: https://bioone.org/journals/American-Fern-Journal on 28 Dec 2023
Terms of Use: https://bioone.org/terms-of-use Access provided by University of Florida

American Fern Journal 113(4):257–262 (2023)
Published on 26 December 2023

Shorter Note

A Genome Size for the Appalachian Gametophyte.—*Vittaria appalachiana* Farrar & Mickel is perhaps the most well-studied fern that exists solely in the gametophyte life stage (reviewed by Farrar, American Journal of Botany 103:593-595. 2016 and Pinson *et al.,* International Journal of Plant Sciences 178:1–18. 2017). The species is distributed widely across the Appalachian mountains in the eastern United States and can form large clonal mats of thousands of gametophytes in protected crevices of non-calcareous rock (Fig. 1). Mature thalli are a single cell thick with a ribbon-like structure that differentiates into two forms: basal branches, which attach to the substrate via rhizoids, and upright branches, which produce gemmae on their margins (Fig. 1C; Farrar and Mickel, American Fern Journal 81:69–75. 1991). While *V. appalachiana* will occasionally produce archegonia on basal branches and antheridia on developing gemmae (Farrar, American Journal of Botany 65:1–12. 1978), sporophytes are almost never produced; the only reported observations are of lab-induced, abortive sporophytes from Caponetti, Whitten, and Beck (American Fern Journal 72: 36–40. 1982), and one documentation of aborted sporophytes found in the field (Farrar, 1978). *Vittaria appalachiana* therefore serves as an excellent model system for understanding prolonged asexual reproduction in the gametophytic life stage, and the breakdown of the transition between gametophyte and sporophyte phases.

Given its unusual habit, *V. appalachiana* has been of great interest to pteridologists, and previous studies have focused on the ecology (Stevens and Emery, Diversity and Distributions 21:1242–1253. 2015; Chambers and Emery, American Journal of Botany 103: 86–98. 2016), physiology (Chambers, Watkins, and Sessa, American Journal of Botany 104:598–607. 2017), phylogenetics (Pinson and Schuettpelz, American Journal of Botany 103:668–676. 2016), population genetics (Farrar, Proceedings of the Royal Society of Edinburgh, 86B: 361–369. 1985; Farrar, Systematic Botany 15:98–111. 1990), and comparative genomics (Pelosi, Barbazuk, and Sessa, International Journal of Plant Sciences, 184:454–469. 2023. https://www.journals.uchicago.edu/doi/full/10.1086/724824) of this species. The base chromosome number in *Vittaria* is x=60, and cytotaxonomic work on *V. appalachiana* recovered a haploid chromosome number of n=120 (Gastony, Systematic Botany 2:43–48. 1977), making it a polyploid relative to its sister species *V. graminifolia* Kaulf. (n=60 [Gastony, 1977], but see Smith and Mickel [Brittonia 29:391-398. 1977] for a count of n=120). However, there is debate about the origin of *V. appalachiana* that largely focuses on the mode of polyploidization. Population genetic results from allozyme studies suggested an allopolyploid origin (Farrar 1985; Farrar, 1990; Farrar, pers. comm.), while a phylogenetic approach pointed to an autopolyploid origin (Pinson and Schuettpelz, 2016).

Despite the clear interest in *V. appalachiana* and the particular attention given to its origin and persistence as a polyploid species, there are, surprisingly,

Downloaded From: https://bioone.org/journals/American-Fern-Journal on 28 Dec 2023
Terms of Use: https://bioone.org/terms-of-use Access provided by University of Florida

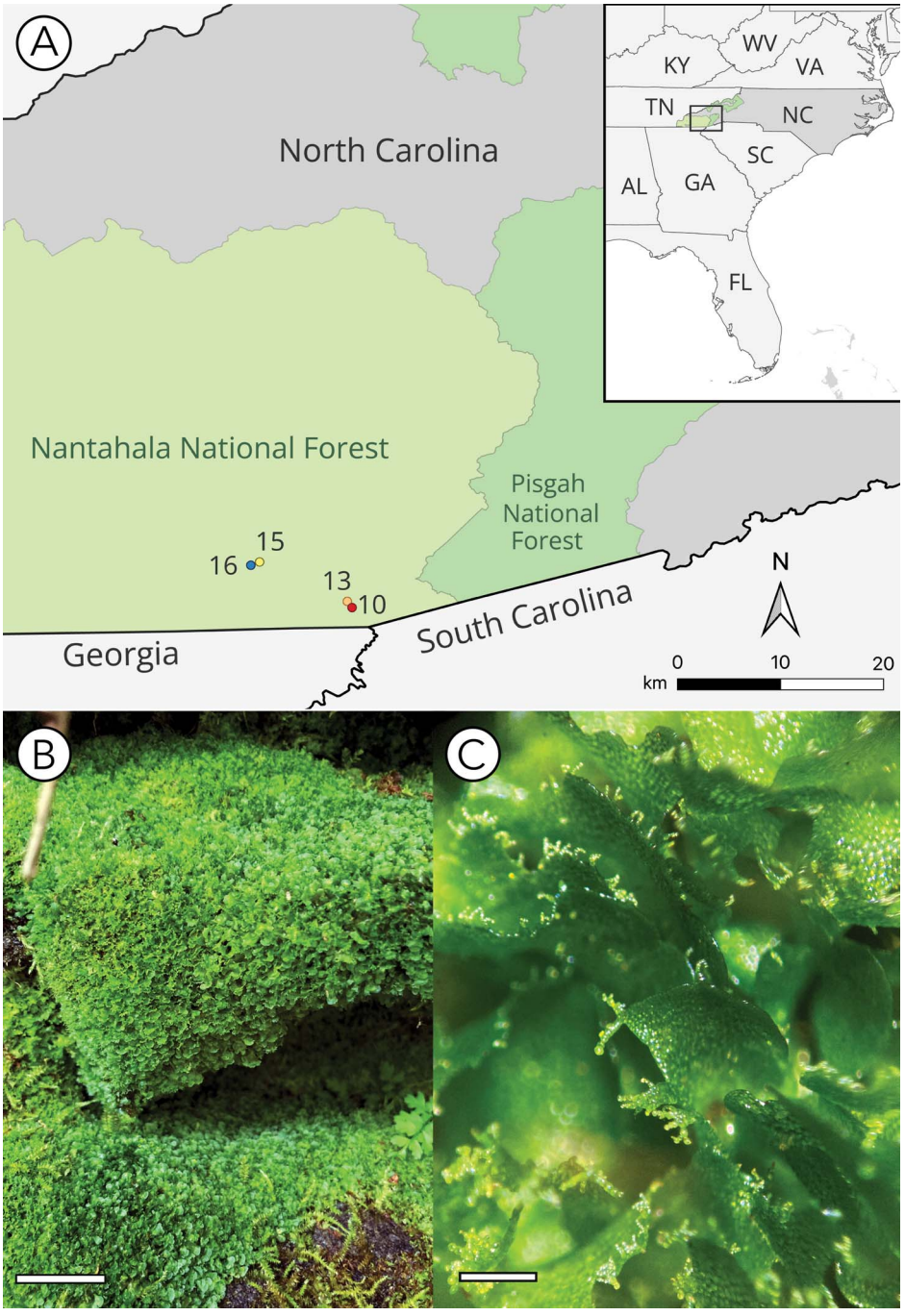


FIG. 1. Collection localities and pictures of *Vittaria appalachiana*. A) Collection map with inset showing the eastern United States. B) *V. appalachiana* at Site 13; scale bar = 10 cm. C) *V. appalachiana* with gemmae on the margin; scale bar = 0.5 mm.

Downloaded From: https://bioone.org/journals/American-Fern-Journal on 28 Dec 2023
Terms of Use: https://bioone.org/terms-of-use Access provided by University of Florida

no estimates available of its genome size. The closest relevant estimate is from *V. lineata* (L.) Sm. (1C=33.92pg, Clark *et al.,* New Phytologist 210:1072–1082. 2016), a sexually-reproducing tetraploid species with the same haploid chromosome number (*n*=120; Gastony, 1977) and that is diverged from *V. appalachiana* by 12-14 MY (Pelosi, Barbazuk, and Sessa, 2023; Pelosi *et al.,* Frontiers in Plant Science 13:882441. 2022). Assessing the genome size of the 'Appalachian gametophyte' would contribute significantly toward a full understanding of this enigmatic fern and evolutionary processes in the genus *Vittaria* as a whole. Genome sizes have further been of particular interest to fern biologists as proxies for ploidal level (*e.g.*, Dauphin, Gran, and Mráz, Plant Systematics and Evolution 302:575-584. 2016; Barrington, Patel, and Southgate, Applications in Plant Sciences e11339. 2020) and have been shown to be correlated with ecological and environmental variables (Wang *et al.,* Plant Diversity 141–152. 2022). A genome size estimate for *V. appalachiana* is the first step in assessing such correlations in this taxon, and here we provide the first such estimate.

Approximately one to two inch square patches of gametophytes were collected from four field sites in North Carolina (Fig. 1A) as part of a larger comparative study (site numbers reported here are kept consistent with that work, in anticipation of its future publication). Sites varied widely in environmental conditions; at site 10, gametophytes were growing on the underside of a large rock overhang in near darkness; at site 13, gametophytes were growing on a vertical rock face receiving ample indirect light (Fig. 1B); at site 15, gametophytes were growing on the rock walls of a shallow cave behind a small waterfall in near darkness; at site 16 gametophytes were growing on the sandy floor of a cave behind a very large waterfall, receiving continuous spray in near darkness. Small mats of gametophytes were carefully removed from larger colonies and stored in petri dishes for transportation. Vouchers were deposited at the New York Botanical Garden's William and Lynda Steere Herbarium (NY). For longer-term cultivation, some of the gametophytes were transferred to large jars containing sphagnum moss and/or rocky media and are maintained as living colonies at the NYBG Pfizer Laboratory.

Portions of each colony were sent to the University of Florida and stored at 4°C prior to preparation for flow cytometry. Any co-occurring bryophytes or other non-fern tissues were removed from samples prior to being chopped for flow cytometry. The absolute genome sizes of gametophytes from the four populations were estimated using the protocol in Pelosi *et al.* (American Fern Journal 113:43–55. 2023). Briefly, three to five gametophytes were pooled and chopped independently of a size standard in woody plant buffer (Loureiro *et al.,* Annals of Botany 100:875-888. 2007) to ensure that nuclei from the sample were obtained. We then co-chopped gametophytes from each population with two internal size standards, A*llium cepa* 'Alice' (2C = 34.89pg), and *Vicia faba* 'Inovec' (2C = 26.90pg; Doležel *et al*., Annals of Botany 82 (Suppl.A):17–26. 1998). Each standard was also run independently for validation purposes in cases where the standard and sample had potentially overlapping peaks due to similarity of genome size. Given that the sample and reference peaks for the

Downloaded From: https://bioone.org/journals/American-Fern-Journal on 28 Dec 2023
Terms of Use: https://bioone.org/terms-of-use Access provided by University of Florida

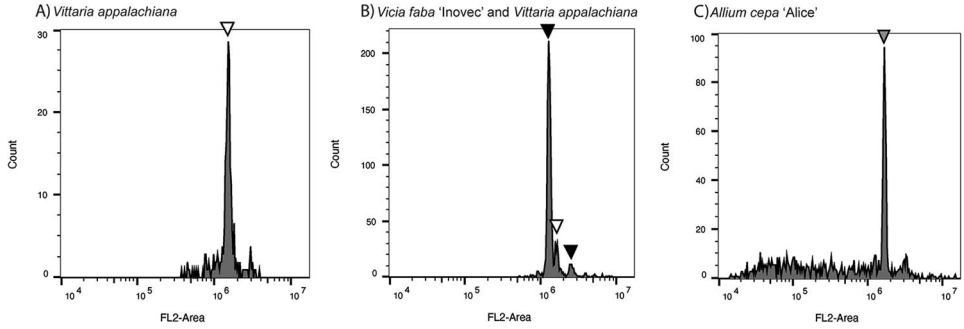


FIG. 2. Examples of histograms showing the relative DNA content of A) *Vittaria appalachiana* (white triangle), B) *Vicia faba* 'Inovec' (black triangles, representing G1 and G2 cell cycle peaks from left to right, respectively) and *V. appalachiana* (white triangle), and C) *Allium cepa* 'Alice' (gray triangle). Both *V. appalachiana* peaks are from sample V.app15 (voucher Sorojsrisom 0147).

samples co-chopped with *Allium* overlapped (Fig. 2A, C), we used this reference as an external standard against the samples run independently of a size standard; *Vicia* was used as an internal standard (Fig. 2B). All specimens were run with both *Allium* and *Vicia* standards, with the exception of the sample from site 16, which only had enough material to run with the *Allium* standard. Samples were stained with PI and analyzed on a BD Accuri C6 flow cytometer (Franklin Lakes, New Jersey, U.S.A.) at the University of Florida ICBR Cytometry Core Facility, RRID:SCR_019119. Data were analyzed using FlowJo ver. 10 (Ashland, Oregon, U.S.A.).

We determined that *V. appalachiana* has a 1C genome size of 33.99 ± 1.57pg (mean ± SD; Table 1). Estimates using *Allium* as the standard were slightly lower (33.01pg) than when *Vicia* was used as the standard (34.98pg), but had similar variances (SD=1.13 and 1.37 for *Allium* and *Vicia,* respectively). This difference may be due to the use of *Allium* as an external standard as the flow cytometer can "drift" from run to run (Doležel *et al.,* 1998). Gametophytes from different populations had relatively consistent genome size estimates (34.21-34.57pg), except for site 15, which had an average 1C estimate of 32.66pg. Differences in the ecology of the sites (*e.g.*, increased water availability at site 15) may influence genome sizes as growth habit has been shown to impact genome size in ferns at a macroevolutionary scale (Wang *et al.,* 2022), although further study of the potential ecological correlates to genome size within this species is required.

The genome size we found for *V. appalachiana* is very close to the estimated genome size of *V. lineata* of 33.92pg (Clark *et al.,* 2016). Lynch (*The Origins of Genome Architecture,* Sinauer Associates. 2007) hypothesized that genome sizes of asexual and selfing species should be greater than their sexual relatives, potentially due to the unrestrained proliferation of transposable elements (TEs). In recent work, Pelosi, Barbazuk, and Sessa (2023) found greater proportions of TEs in the transcriptomes of *V. appalachiana* compared to *V. lineata*, yet here we find that the estimated genome sizes are nearly identical.

Downloaded From: https://bioone.org/journals/American-Fern-Journal on 28 Dec 2023
Terms of Use: https://bioone.org/terms-of-use Access provided by University of Florida

TABLE 1. Collection and voucher information for specimens used in this study and associated genome size estimates.

Sample	Voucher	Latitude	Longitude	Population	Size Standard	Type of Standard	1C Genome Size (pg)
V.app10	Sorojsrisom 0092	35.022328	-83.126725	10	*Allium cepa* 'Alice'	External	33.21
V.app10	Sorojsrisom 0092	35.022328	-83.126725	10	*Vicia faba* 'Inovec'	Internal	35.21
V.app13	Sorojsrisom 0099	35.029167	-83.131944	13	*Allium cepa* 'Alice'	External	34.47
V.app13	Sorojsrisom 0099	35.029167	-83.131944	13	*Allium cepa* 'Alice'	External	32.52
V.app13	Sorojsrisom 0099	35.029167	-83.131944	13	*Vicia faba* 'Inovec'	Internal	36.74
V.app16	Sorojsrisom 0148	35.069167	-83.238611	16	*Allium cepa* 'Alice'	External	34.47
V.app15	Sorojsrisom 0147	35.072778	-83.22896	15	*Allium cepa* 'Alice'	External	31.82
V.app15	Sorojsrisom 0147	35.072778	-83.22896	15	*Vicia faba* 'Inovec'	Internal	33.49

However, differences in genome size may evolve slowly and may not yet be apparent due to the recent origin of asexual reproduction in *V. appalachiana* (estimated between 2 mya and 130 kya; Farrar, 1990). The four-fold greater genome sizes of *V. appalachiana* and *V. lineata* relative to that of their sister genus *Adiantum* (between 5.13pg and 7.81pg; Bainard *et al.,* Chromosome Research 19:763. 2011, Clark *et al.* 2016), correspond to the two rounds of whole genome duplication (WGD) in the history of the *Vittaria* species suggested by Pelosi *et al.* (2022) and Pelosi, Barbazuk, and Sessa (2023): one WGD at the base of the vittarioid ferns (PTER3 *sensu* Pelosi *et al.,* 2022) and a second round of duplications that occurred independently in the *V. lineata* and *V. appalachiana* lineages and are unshared between the two species.

More extensive cytological and phylogenetic study on *Vittaria* will be required to determine the extent and mode(s) of polyploidization among its species, especially in *V. appalachiana*. Additional data on the correlates of genome size in this gametophyte-only fern will continue to improve our understanding of its evolution, and we hope that the genome size estimate provided for *V. appalachiana* here will serve as a foundation for these studies.

ACKNOWLEDGEMENTS

The authors are continuously inspired by the foundational work of Don Farrar on gametophyte biology and on *Vittaria appalachiana* in particular. Funding for this project was provided by the Southern Appalachian Botanical Society Earl Core Student Research Award to J.A.P., Torrey Botanical Society Graduate Student Research Fellowship to B.A.Z., Maxwell-Hanrahan Field Research Fund and Highlands Biological Station Grants-In-Aid of Research to E.S.S. We thank

Downloaded From: https://bioone.org/journals/American-Fern-Journal on 28 Dec 2023
Terms of Use: https://bioone.org/terms-of-use Access provided by University of Florida

W. Brad Barbazuk and Ruth Davenport for use of lab space. Samples were collected with permission from the USDA Forest Service.—Jessie A. Pelosi[1,*], Bethany A. Zumwalde[1], Elissa S. Sorojsrisom[2,3], and Emily B. Sessa[1,3]. [1]Department of Biology, University of Florida, Gainesville, FL 32611, [2]Department of Ecology, Evolution, and Environmental Biology, Columbia University, NY, 10027, [3]William and Lynda Steere Herbarium, The New York Botanical Garden, Bronx, NY 10458, *author for correspondence, jessiepelosi@ufl.edu.

Downloaded From: https://bioone.org/journals/American-Fern-Journal on 28 Dec 2023
Terms of Use: https://bioone.org/terms-of-use Access provided by University of Florida